

實習生月報

PDO 資料路徑重建

CoE / SDO 交易重建

PDO Mapping / Assignment 關聯整理

SyncManager / FMMU 資料路徑分析

報告期間

2026.08

MONTHLY REPORT

本月工作大綱



7/30 - 8/5

理解封包與 EtherCAT 100%

- 使用 Wireshark 觀察從上電抓的原始 EtherCAT 封包
- 整理 Broadcast、AP、FP 等指令對 Slave 的操作方式
- 整理 Master 對 Slave ESC 暫存器的讀寫流程
- 理解 CoE、SDO、PDO 與 Object Dictionary 的關係
- 建立本月主線：CoE / SDO → PDO Mapping / Assignment → SyncManager → FMMU

8/6 - 8/12

建立 Slave 拓樸關係 80%

- 理解 Master 如何透過 Auto Increment / Configured Address 指定 Slave
- 理解 Configured Station Address 與 Configured Station Alias 的差異
- 設計 AP Datagram 配對、Slave Discovery 模型與分析流程

設計 Analyzer 驗證 Agent 20%

- 建立 Stage 3 Result Check
- 由 Capture evidence 驗證 Slave Count、Topology、Address 與 Identity

8/13 - 8/19

Mailbox / CoE / SDO 70%

- 在 Datagram Record 中加入 CoE / SDO decoded fields
- 重建 SDO Initiate Download Request / Response Transaction
- 依 SDO Index / SubIndex 辨識 PDO Mapping / Assignment 操作
- 重建 RxPDO / TxPDO Mapping Configuration
- 重建 0x1C12 / 0x1C13 PDO Assignment Configuration

完善 Agent 驗證流程 30%

- 整理 TShark EtherCAT / CoE / SDO 查詢與 JSON 結構規格
- 將常用 Capture 查詢封裝成 Python tools，提供 LLM 多種驗證方式

8/20 - 8/21

SyncManager / FMMU 分析 100%

- 在 Datagram Record 中加入 SyncManager / FMMU decoded fields
- 重建 SyncManager Configuration
- 重建 FMMU Configuration
- 驗證 PDO Assignment 與 SM2 / SM3 的設定關係
- 驗證 FMMU Physical Address 與 SyncManager RAM 區段的對應

SECTION / 01

01

上電流程與 ESC Register 解析

追蹤啟動封包，建立 ESC 暫存器存取與解析脈絡

上電封包操作流程

01 Broadcast Initialization

BRD / BWR / BRW
All Slaves

02 Auto Increment Addressing

APRD / APWR / APRW
Topology Position

03 Configured Station Address

APWR / ADO 0x0010
Fixed Address

04 Fixed Position ESC Access

FPRD / FPWR / FPRW
Configured Address + ADO

Master

Slave 1

Outgoing ADP: 0x0000
Signed: 0
Topology Position: 1

Slave 2

Outgoing ADP: 0xFFFF
Signed: -1
Topology Position: 2

Slave 3

Outgoing ADP: 0xFFFE
Signed: -2
Topology Position: 3

Outgoing ADP

Signed Value

命中的拓樸位置

0x0000

0

Slave 1

0xFFFF

-1

Slave 2

0xFFFE

-2

Slave 3

Master 以 outgoing ADP = 0、-1、-2... 逐站指定 Slave。
ADO 是目標 ESC Register Offset，命中的 Slave 會執行該 Datagram。

SII EEPROM 讀取流程

01 檢查 EEPROM 狀態

- 讀取 ESC Register **0x0502~0x0503**
- 確認 Busy 清除且沒有 Error

02 指定 SII Word Address

- 寫入 ESC Register **0x0504~0x0507**
- **0x0008** = Vendor ID · **0x000A** = Product Code 起始 Word

03 送出 EEPROM Read Command

- 在 **0x0502~0x0503** 的 Control / Status 欄位送出 Read Command

04 讀取 EEPROM Data

- 以 APRD 讀取 **ADO = 0x0508**
- 依前面指定的 SII Word Address 解讀資料內容

Address Space	Address	Meaning
ESC Register	0x0502~0x0503	EEPROM Control / Status
ESC Register	0x0504~0x0507	EEPROM Address
ESC Register	0x0508~0x050F	EEPROM Data
SII EEPROM Word	0x0008~0x0009	Vendor ID
SII EEPROM Word	0x000A~0x000B	Product Code

實際 Capture

實際 Capture 中，Master 會以一個 **APWR** Datagram，從 **ADO = 0x0502** 開始一次寫入 **6 Bytes**。前 2 Bytes 對應 **0x0502~0x0503** 的 EEPROM Control / Status，後 4 Bytes 對應 **0x0504~0x0507** 的 EEPROM Address，因此一次 **APWR** 即可同時送出 Read Command 與欲讀取的 SII Word Address。

EEPROM Read 完成後，Master 再以 **APRD** 讀取 **ADO = 0x0508** 的 EEPROM Data。實際取得的資料內容，則由前面寫入的 SII Word Address 決定，例如 **0x0008** 對應 Vendor ID、**0x000A** 對應 Product Code。



Slave Discovery 重建成果

01 重點統整

AP Commands

02 AP-Analyzer 重建流程

Slave Discovery

03 程式碼實作

Class Relationship

04 實際成果

Stage 3 Result

Slave Discovery

| AP Datagram

| Topology Position

| Configured Address

| Slave Identity

SECTION / 02

02

CoE / SDO 與 PDO Mapping 重建

從 CoE / SDO Transaction 重建 PDO Mapping 與 Assignment，建立 Process Data 的資料定義。

CoE / SDO 知識整理

01 Mailbox / CoE / SDO
Protocol Layers

02 Object Dictionary
Index / SubIndex

03 SDO Download / Upload
Client / Server

04 SDO → PDO
Configuration
PDO Mapping

CoE / SDO 知識主線

EtherCAT Datagram

Mailbox

CoE

SDO

Object Dictionary

PDO Configuration

CoE / SDO 的作用是透過 EtherCAT Mailbox 存取 CANopen Object Dictionary，並可進一步用來設定 PDO Mapping / Assignment。

SDO Transaction / PDO Mapping 重建成果

EtherCATAnalyzer.TestConsole.exe analyze

Stage 4 / 105 Transactions

[Stage 4] SDO Transaction Analysis

Elapsed: 6.226 s

Transaction Count: 105

Configured Slave Address:	0x0001,	Object:	0x1C12:00,	Data:	0x00,	Request Frame:	41394,	Response Frame:	41403,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1C13:00,	Data:	0x00,	Request Frame:	41408,	Response Frame:	41417,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A00:00,	Data:	0x00,	Request Frame:	41422,	Response Frame:	41455,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A00:01,	Data:	0x60410010,	Request Frame:	41460,	Response Frame:	41614,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A00:02,	Data:	0x60430010,	Request Frame:	41639,	Response Frame:	41786,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A00:03,	Data:	0x30221110,	Request Frame:	41791,	Response Frame:	41958,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A00:04,	Data:	0x30210c10,	Request Frame:	41963,	Response Frame:	42130,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A00:05,	Data:	0x30210210,	Request Frame:	42135,	Response Frame:	42302,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A00:00,	Data:	0x05,	Request Frame:	42307,	Response Frame:	42316,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A01:00,	Data:	0x00,	Request Frame:	42321,	Response Frame:	42330,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A02:00,	Data:	0x00,	Request Frame:	42335,	Response Frame:	42344,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1A03:00,	Data:	0x00,	Request Frame:	42349,	Response Frame:	42376,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1600:00,	Data:	0x00,	Request Frame:	42381,	Response Frame:	42496,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1600:01,	Data:	0x60420010,	Request Frame:	42501,	Response Frame:	42676,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1600:00,	Data:	0x01,	Request Frame:	42681,	Response Frame:	42690,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1601:00,	Data:	0x00,	Request Frame:	42695,	Response Frame:	42704,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1601:01,	Data:	0x60400010,	Request Frame:	42709,	Response Frame:	42842,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1601:02,	Data:	0x60600008,	Request Frame:	42847,	Response Frame:	43014,	Success:	True,	Abort Code:	N/A
Configured Slave Address:	0x0001,	Object:	0x1601:03,	Data:	0x60720010,	Request Frame:	43019,	Response Frame:	43193,	Success:	True,	Abort Code:	N/A



SDO / PDO Analyzer 實作設計

01 設計流程

Reconstruction Pipeline

02 程式碼關係

Class Relationship

SDO Transaction

PDO Mapping

PDO Assignment



SECTION / 03

03

SyncManager 與 Local RAM 資料交換

解析 SyncManager Configuration，確認 PDO Process Data 實際使用的 ESC Local RAM Buffer。

SyncManager 功能與 PDO Assignment 關係

01

SyncManager

Local RAM Control

02

RxPDO Path

0x1C12 → SM2

03

TxPDO Path

0x1C13 → SM3

PDO Mapping



PDO Assignment



SyncManager



ESC Local RAM

0x1C12 / 0x1C13 決定目前使用哪些 PDO Mapping ;
SyncManager 則管理 Process Data 在 ESC Local RAM 中的實際資料區域與存取方式。



SyncManager Analyzer 實作設計

01 設計流程
Reconstruction Pipeline

02 程式碼關係
Class Relationship

SyncManager Configuration Reconstruction

SM Register → SyncManager Configuration → Local RAM Layout

SyncManager Analyzer 重建成果

[PDO Assignment Summary]

Station Address: 0x0001
Assignment Object: 0x1C12
Type: RxPDO Assignment
Active Entries: 1

Entry	Mapping Object
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01	0x1600
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Station Address: 0x0001
Assignment Object: 0x1C13
Type: TxPDO Assignment
Active Entries: 1

Entry	Mapping Object
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01	0x1A00
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[SyncManager Summary]

Station Address: 0x0001
SyncManager: SM2



SECTION / 04

04

FMMU 與 Logical Address Mapping

解析 FMMU Configuration，建立 EtherCAT Logical Address 與 ESC Physical RAM 的映射關係。

FMMU Analyzer 實作設計

01

FMMU

Logical Mapping

02

設計流程

Reconstruction Pipeline

03

程式碼關係

Class Relationship

04

重建成果

Analyzer Result

FMMU (Fieldbus Memory Management Unit) 是 ESC 內的硬體單元，負責把 EtherCAT Logical Address 空間映射到 ESC 的 Physical / Local RAM 區域。Master 因此能以連續邏輯位址讀寫 Process Data，不必直接處理各 Slave 內部 RAM 的實體位址。

PDO Assignment 決定使用哪些 PDO Mapping；SyncManager 管理這批 Process Data 實際使用的 Local RAM Buffer；FMMU 再把該 RAM 區域映射到 EtherCAT Logical Address。

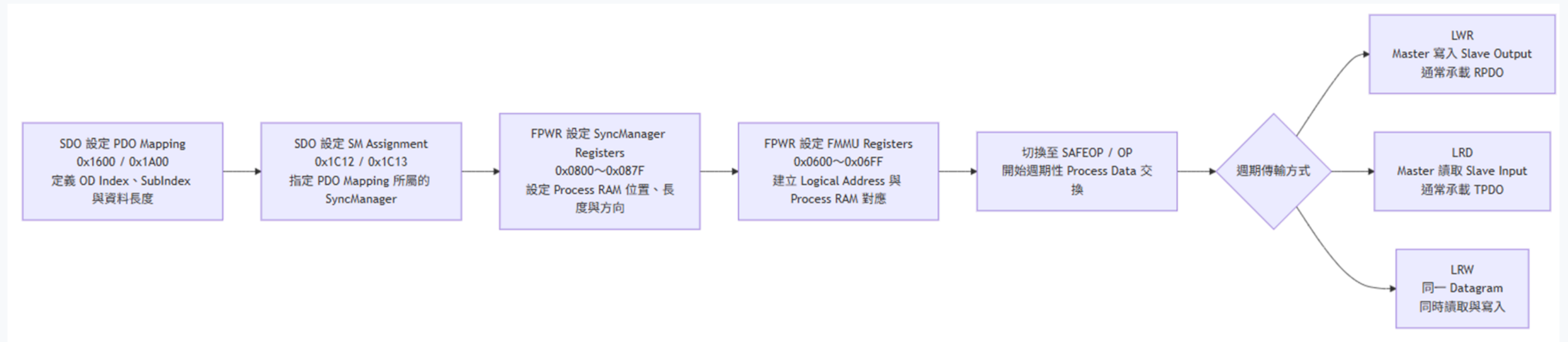
PDO Mapping / Assignment → SyncManager → FMMU → Process Data

Field	Meaning
Logical Start Address	EtherCAT logical address space 的映射起點
Length	此 FMMU mapping 涵蓋的資料長度
Physical Start Address	對應 ESC Local RAM 的起始位址
Direction / Orientation	定義 logical read / write 與 local access 的方向
Enabled / Activate	啟用或停用此 FMMU mapping

SyncManager 管理實體 Local RAM，FMMU 管理 Logical Address 到 Local RAM 的映射。兩者串接後，Master 才能透過 LRW、LRD、LWR 等 Process Data 命令操作 PDO 資料。



總結



EtherCAT 圖解分析

FIGURE / 04



[Diagram Canvas]

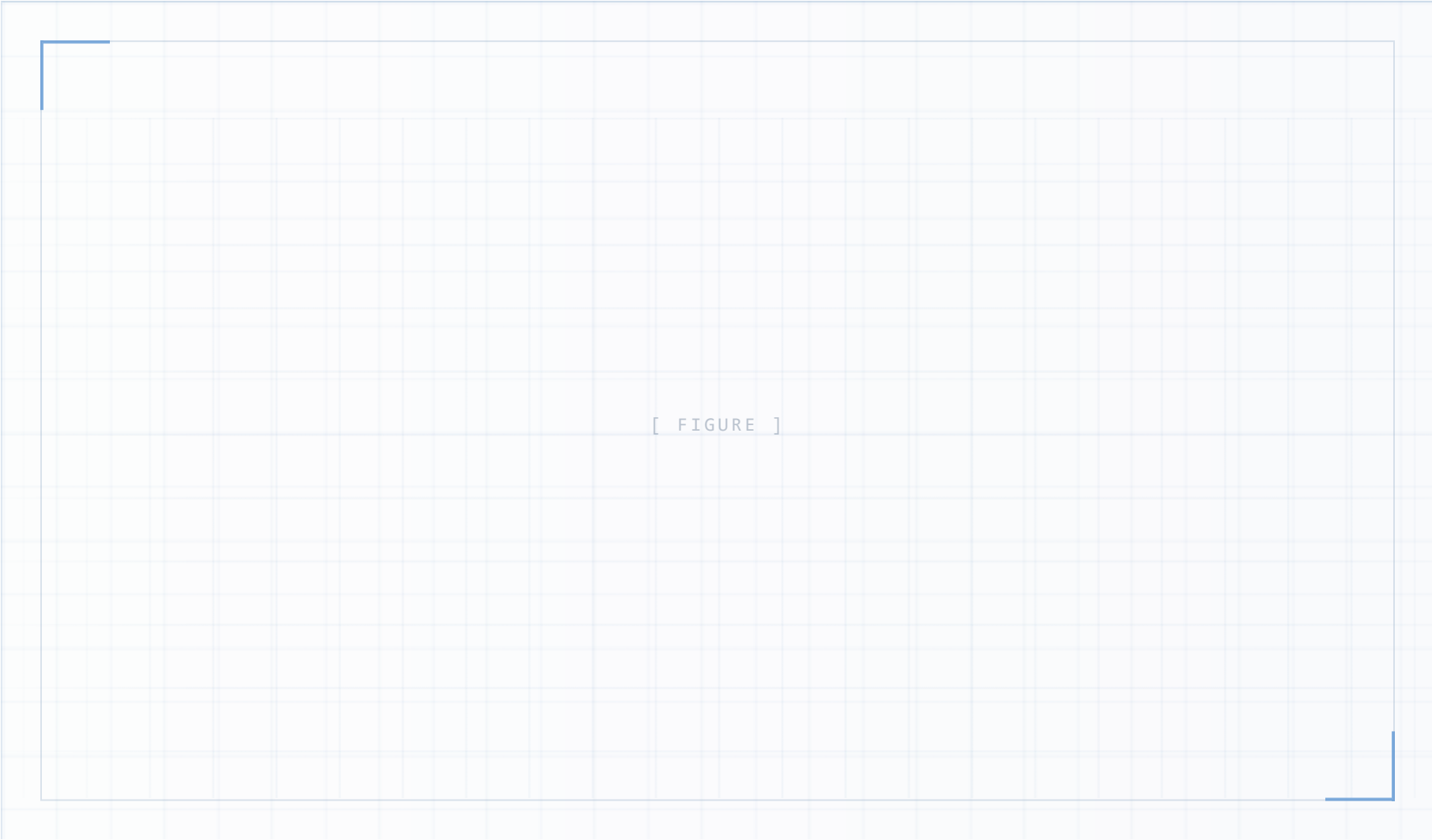
Class Diagram / Architecture Diagram / Data Path Figure

REPLACE WITH IMG / SVG / FIGURE / HTML DIAGRAM

FOCUS POINTS

- 01 Diagram Purpose**
Explain the main relationship between modules.
- 02 Key Components**
Highlight the major classes or architecture blocks.
- 03 Reading Order**
Describe how to read the diagram efficiently.
- 04 Implementation Relevance**
Connect the diagram to the current development work.

EtherCAT 資料路徑概覽



KNOWLEDGE POINTS

01

FMMU

Logical Address
→ ESC local

02

SyncManager

Consistent data
exchange control

03

Process RAM

Stores mapped
process data

04

DATA PATH

Logical Address → FMMU →
SM → RAM → Application

